



Beata Zalewa

**Tworzenie Inteligentnych Asystentów
z Azure OpenAI Assistants API (Preview)**



**Pytech
Summit**

All about Python **2024**

10

Grudnia 2024



0 mnie



Security Architect



Consultant



Microsoft Certified Trainer



AI & Cybersecurity Practitioner



Developer



Freelancer



Azure @ ❤️



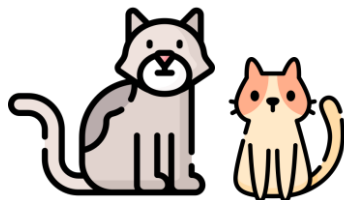
Google Cloud



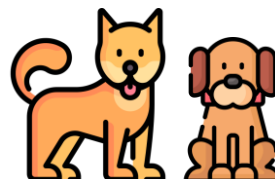
1 Mąż



1 Córka



2 Koty



2 Psy



Kryminały



Fotografia

Agenda

- Wprowadzenie do Azure OpenAI Assistants API
- Tworzenie asystentów w Azure AI Foundry
- Tworzenie asystentów w kodzie
- Sesja Q&A

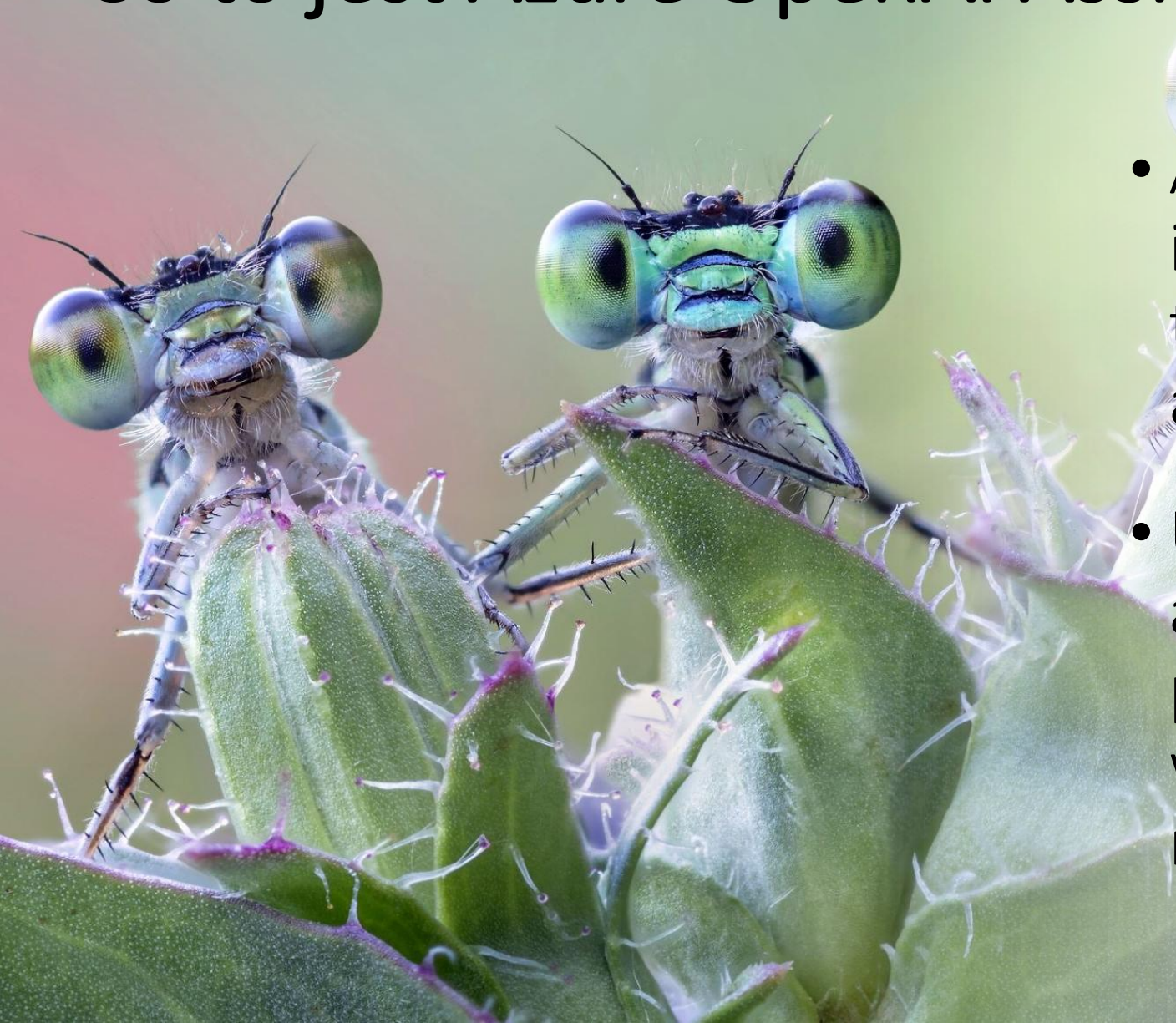




Zaczynamy!

Co to jest Azure OpenAI Assistants API?

- Azure OpenAI Assistants API to interfejs, który umożliwia tworzenie inteligentnych asystentów AI w aplikacjach.
- Umożliwia dostosowanie asystentów do specyficznych potrzeb użytkowników, wykorzystując zaawansowane modele AI.



Kluczowe funkcje API

- **Personalizacja:** Asystenci mogą być dostosowywani za pomocą niestandardowych instrukcji.
- **Integracja narzędzi:** Obsługuje różne narzędzia, takie jak interpreter kodu i wyszukiwanie plików.
- **Zarządzanie wątkami:** Umożliwia przechowywanie historii rozmów i zarządzanie kontekstem.

Instructions ⓘ

You are a helpful AI assistant who makes interesting visualizations based on data. You have access to a sandboxed environment for writing and testing code. When you are asked to create a visualization you should follow these steps:

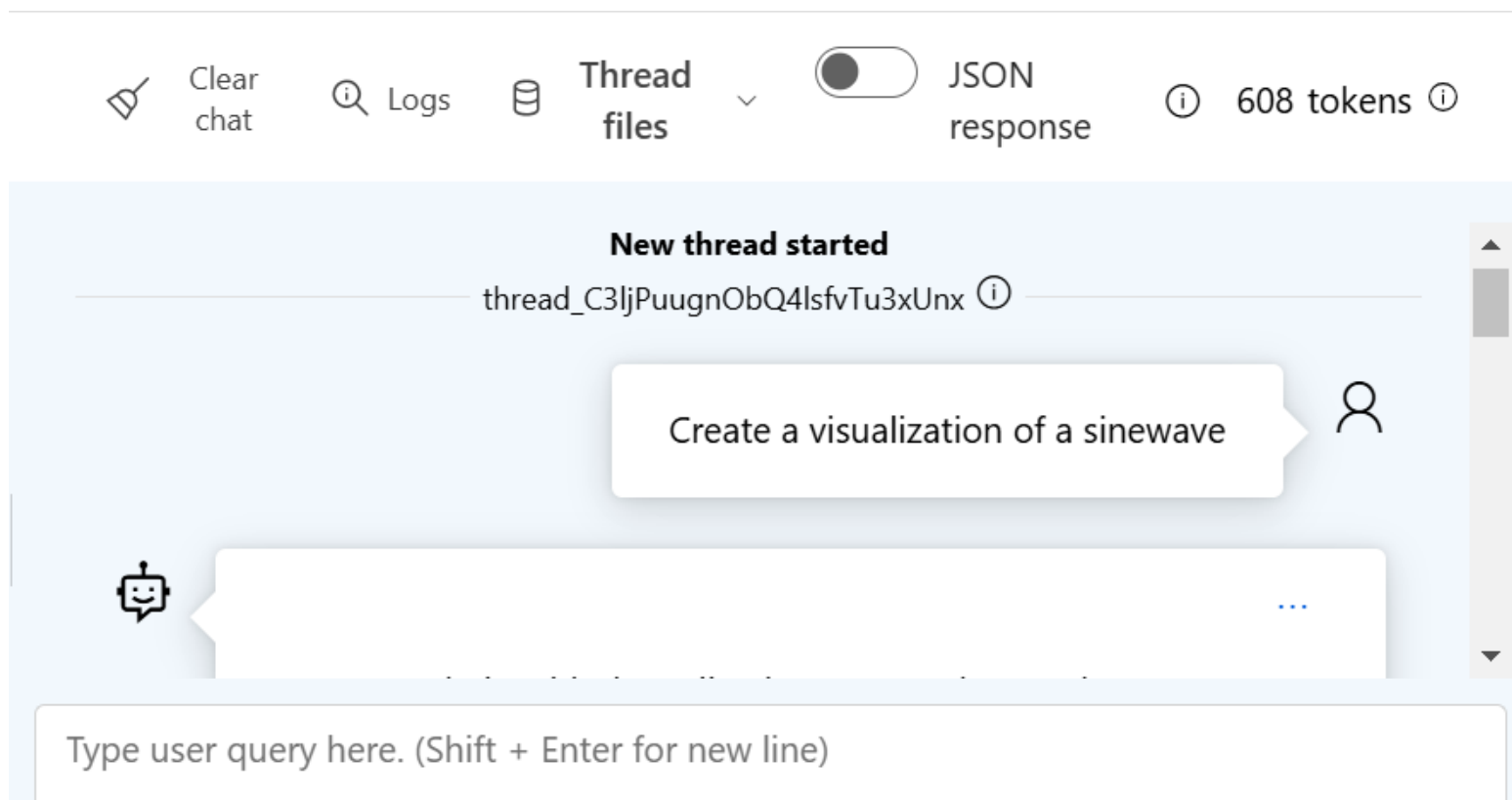
1. Write the code.
2. Anytime you write new code display a preview of the code to show your work.
3. Run the code to confirm that it runs.
4. If the code is successful display the visualization
5. If the code is unsuccessful display the error message and try to revise the code and rerun going through the steps from above again.

Jak działa API ?

- Asystenci korzystają z modeli OpenAI, które wykonują zadania na podstawie przekazanych instrukcji.
- Można tworzyć wątki konwersacji, które przechowują wiadomości i ułatwiają interakcję z użytkownikami.

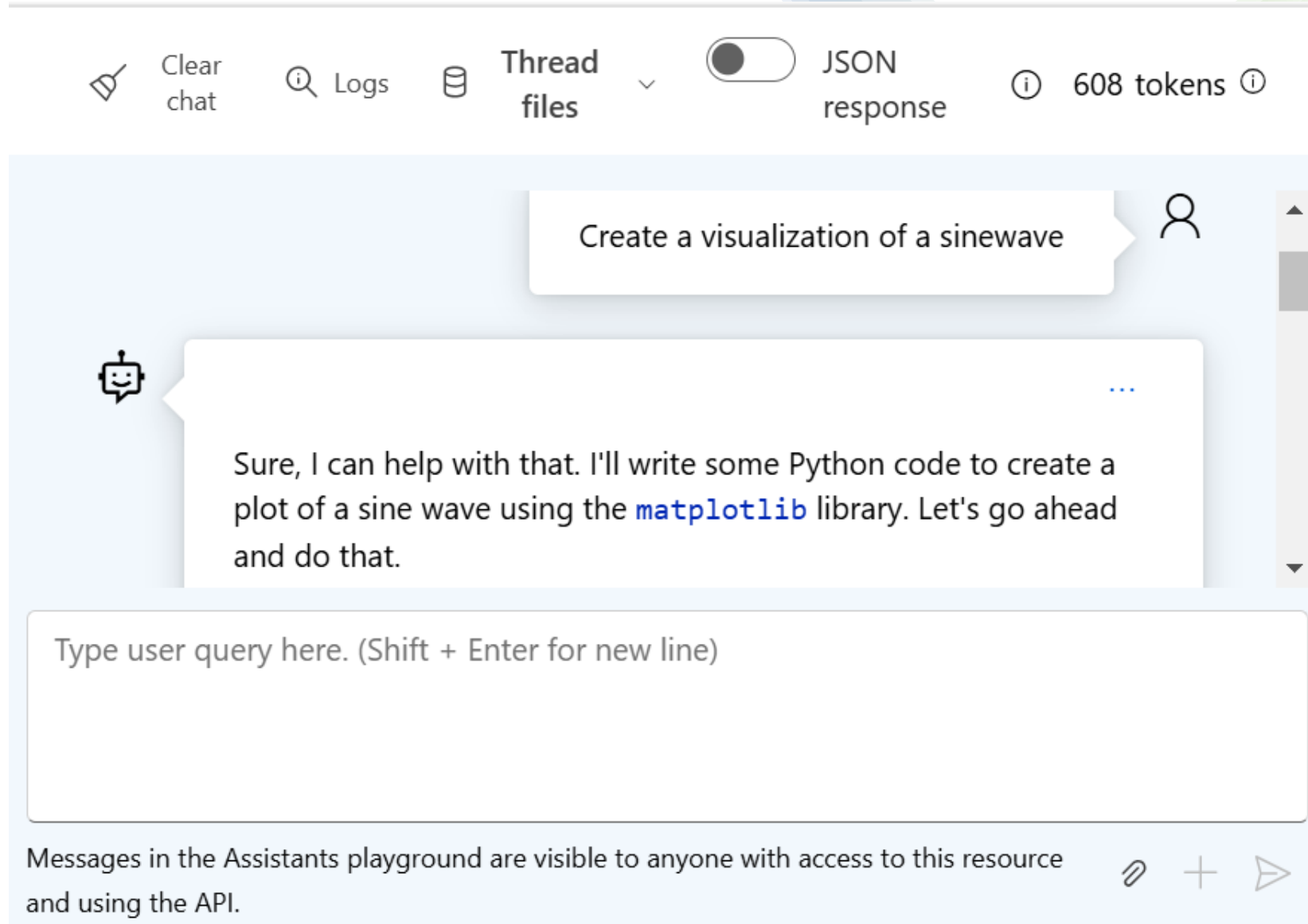
Składniki asystentów

- **Asystent:** Niestandardowy projekt AI korzystający z modeli usługi Azure OpenAI w połączeniu z narzędziami.
- **Wątek (Thread):** Sesja konwersacji między asystentem a użytkownikiem. Wątki przechowują komunikaty i automatycznie obsługują obcinanie w celu dopasowania zawartości do kontekstu modelu.



Składniki asystentów

- **Wiadomość (Message):**
Komunikat utworzony przez Asystenta lub użytkownika. Wiadomości mogą zawierać tekst, obrazy i inne pliki. Komunikaty są przechowywane jako lista w wątku.



Składniki asystentów

- **Uruchomienie (Run):** Aktywacja asystenta w celu rozpoczęcia działania na podstawie zawartości wątku. Asystent używa konfiguracji i komunikatów wątku do wykonywania zadań przez wywoływanie modeli i narzędzi. W ramach przebiegu Asystent dołącza komunikaty do wątku.
- **Krok uruchamiania (Run step):** Szczegółowa lista kroków, które asystent wykonał w ramach przebiegu. Asystent może wywoływać narzędzia lub tworzyć komunikaty podczas jego uruchamiania. Badanie kroków uruchamiania pozwala zrozumieć, jak asystent uzyskuje wyniki końcowe.

Create Azure OpenAI ...

[Learn more](#)

Project Details

Subscription * ⓘ

ZALNET-PAYG

Resource group * ⓘ

(New) PytechSummit2024RG

[Create new](#)

Instance Details

Region ⓘ

East US

Name * ⓘ

pytechsummit2024demo

Pricing tier * ⓘ

Standard S0

[View full pricing details](#)



[Home](#) > [Create a resource](#) > [Marketplace](#) >

Create Azure OpenAI ...

- Basics
- 2** Network
- 3 Tags
- 4 Review + submit

Configure network security for your Azure AI services resource.

Type *

- ☒ All networks, including the internet, can access this resource.
- ☐ Selected networks, configure network security for your Azure AI services resource.
- ☐ Disabled, no networks can access this resource. You could configure private endpoint connections that will be the exclusive way to access this resource.

Create Azure OpenAI ...

- ✔ Basics
- ✔ Network
- ✔ Tags
- 4** Review + submit

🔗 [View automation template](#)

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Basics

Subscription	ZALNET-PAYG
Resource group	PytechSummit2024RG
Region	East US



**pytechsummit2024demo**



Azure OpenAI

Search



Go to Azure AI Foundry portal



Delete



Overview



Activity log



Access control (IAM)



Tags



Diagnose and solve problems



Resource Management



Keys and Endpoint



Encryption



Pricing tier



Networking



Identity



Cost analysis

Get Started

Develop

Monitor

Build your own secure copilot and generative AI applications with Azure OpenAI Service

Deploy an Azure OpenAI model and start making API calls. Connect your own data, call functions, and improve workflow with Azure OpenAI language, image and speech models. You can access the service through REST APIs, Python SDK, or our web-based interface in the Azure AI Foundry portal.

[Learn More](#)



Explore and deploy

Explore and deploy the generative AI models, craft unique prompts for your use cases, and fine-tune select models.

Explore Azure AI Foundry portal



Assistants playground

🔍 Help

🏠 Home

Get started

📦 Model catalog

Playgrounds ^

💬 Chat

🤖 Assistants PREVIEW

🎙️ Real-time audio PREVIEW

🖼️ Images

🗨️ Completions

Tools ^

🧪 Fine-tuning

🔗 Azure OpenAI PREVIEW

Evaluation



Create an assistant

to get started with the [Assistants API](#)

📘 Note: In order to use the Assistants playground you need to deploy a model that supports Assistants. [Learn more about supported models](#) 🔗

Create new deployment

+ Create an assistant

Select a chat completion model

Choose a model to create a new deployment. For flows and other resources, create a deployment from their respective list. [Go to model catalog.](#)

Models: 4

Inference tasks: Chat completion



Show description

Search



gpt-4o

Chat completion



gpt-4o-mini

Chat completion



gpt-4

Chat completion



gpt-35-turbo

Chat completion



Task: Chat completion

GPT-4o offers a shift in how AI models interact with multimodal inputs. By seamlessly combining text, images, and audio, GPT-4o provides a richer, more engaging user experience.

Matching the intelligence of GPT-4 Turbo, it is remarkably more efficient, delivering text at twice the speed and at half the cost. Additionally, GPT-4o exhibits the highest vision performance and excels in non-English languages compared to previous OpenAI models.

GPT-4o is engineered for speed and efficiency. Its advanced ability to handle complex queries with minimal resources can translate into cost savings and performance.

The introduction of GPT-4o opens numerous possibilities for businesses in various

Confirm

Cancel

- Home
- Get started
- Model catalog
- Playgrounds
- Chat
- Assistants** PREVIEW
- Real-time audio PREVIEW
- Images
- Completions
- Tools
- Fine-tuning
- Azure OpenAI PREVIEW
- Evaluation
- Batch jobs

Deploy model gpt-4o

Deployment name *

gpt-4o

Deployment type

Standard

Standard: Pay per API call with lower rate limits. Adheres to Azure data residency promises. Best for intermittent workloads with low to medium volume. Learn more about [Standard deployments](#).

Deployment details

Collapse

Model version

2024-05-13 (Default)

AI resource

pytechsummit2024demo

150K tokens per minute quota available for your deployment

Tokens per Minute Rate Limit



pytechsummit2024demo

(eastus, S0)



Help

s. [Learn more about](#)

new deployment

-  Home
- Get started
-  Model catalog
- Playgrounds ^
-  Chat
-  Assistants PREVIEW
-  Real-time audio PREVIEW
-  Images
-  Completions
- Tools ^
-  Fine-tuning
- Azure
-  OpenAI PREVIEW
- Evaluation
-  Batch jobs
- Shared resources ^

Assistants playground

 Help

 New assistant

 Select assistant

 Delete

 View code

Setup

 Hide

Assistant id: ⓘ

asst_tiRaX141e9CiEnX0emuGBbs

Assistant name ⓘ

Assistant560

Assistant id: asst_tiRaX141e9CiEnX0emuGBbs

Deployment * [+ Create new deployment](#) v

gpt-4o (version:2024-05-13)

Instructions ⓘ

You are a helpful AI assistant who makes interesting visualizations based on data. You have access to a sandboxed environment for writing and testing code. When you are asked to create a should follow these steps: ✓ Last saved: today 7:07 AM

 Clear chat

 Logs

 Thread files v

☒ JSON response

ⓘ 0 tokens ⓘ

New thread started
thread_XP8W8pMbGFrLy7WHbh1Kbfnw ⓘ

Type user query here. (Shift + Enter for new line)

Messages in the Assistants playground are visible to anyone with access to this resource and using the API.   



Assistants playground

🔍 Help

- 🏠 Home
- Get started
- 📦 Model catalog
- Playgrounds ▴
- 💬 Chat
- 🤖 Assistants** PREVIEW
- 🔊 Real-time audio PREVIEW
- 🖼️ Images
- 📄 Completions
- Tools ▴
- 🔧 Fine-tuning
- Azure
- 🔧 OpenAI PREVIEW
- Evaluation



Create an assistant

to get started with the [Assistants API](#)

Deployment

gpt-4o ▾

+ Create an assistant



Home

Get started

Model catalog

Playgrounds 

Chat

 Assistants PREVIEW

 Real-time audio PREVIEW

 Images

 Completions

Tools 

 Fine-tuning

 Azure OpenAI Evaluation PREVIEW

 Batch jobs

Assistants playground

 Help

 New assistant

 Select assistant

 Delete

 View code

 Clear chat

 Logs

 Thread files 



JSON response 

0 tokens 

New thread started

thread_XoSL0ypZ2oyAK76JcqygZMHQ 

Type user query here. (Shift + Enter for new line)

Tools

✓ Last saved: today 6:59 AM

Messages in the Assistants playground are visible to anyone with access to this resource and using the API.

- 
-  Home
 - Get started
 -  Model catalog
 - Playgrounds 
 -  Chat
 -  Assistants PREVIEW
 -  Real-time audio PREVIEW
 -  Images
 -  Completions
 - Tools 
 -  Fine-tuning
 - Azure
 -  OpenAI PREVIEW
 - Evaluation
 -  Batch jobs

Assistants playground

 Help

 New assistant  Select assistant  Delete  View code

 Tools



File search 

 Add vector store



Code interpreter 

 Add files

Functions 

 Add function

 Model settings

✓ Last saved: today 7:00 AM

 Clear chat  Logs  Thread files   JSON response  0 tokens 

New thread started

thread_tgUP3x9aftiHvNhjVOWWIDXY 

Type user query here. (Shift + Enter for new line)

Messages in the Assistants playground are visible to anyone with access to this resource and using the API.   

-  Chat
-  Assistants PREVIEW
-  Real-time audio PREVIEW
-  Images
-  Completions
- Tools
 -  Fine-tuning
 -  Azure OpenAI Evaluation PREVIEW
 -  Batch jobs
- Shared resources
 -  Deployments
 -  Quota
 -  Safety + security
 -  Data files
 -  Vector stores PREVIEW

Assistants playground

 Help

 New assistant  Select assistant  Delete  View code

 Show setup  Clear chat  Logs  Thread files  JSON response  608 tokens 

New thread started

thread_C3ljPuugnObQ4lsfvTu3xUnx 

Create a visualization of a sinewave 



Sure, I can help with that. I'll write some Python code to create a plot of a sine wave using the `matplotlib` library. Let's go ahead and do that.

Type user query here. (Shift + Enter for new line)



Chat

Assistants PREVIEW

Real-time audio PREVIEW

Images

Completions

Tools ^

Fine-tuning

Azure OpenAI PREVIEW
Evaluation

Batch jobs

Shared resources ^

Deployments

Quota

Safety + security

Data files

Vector stores PREVIEW

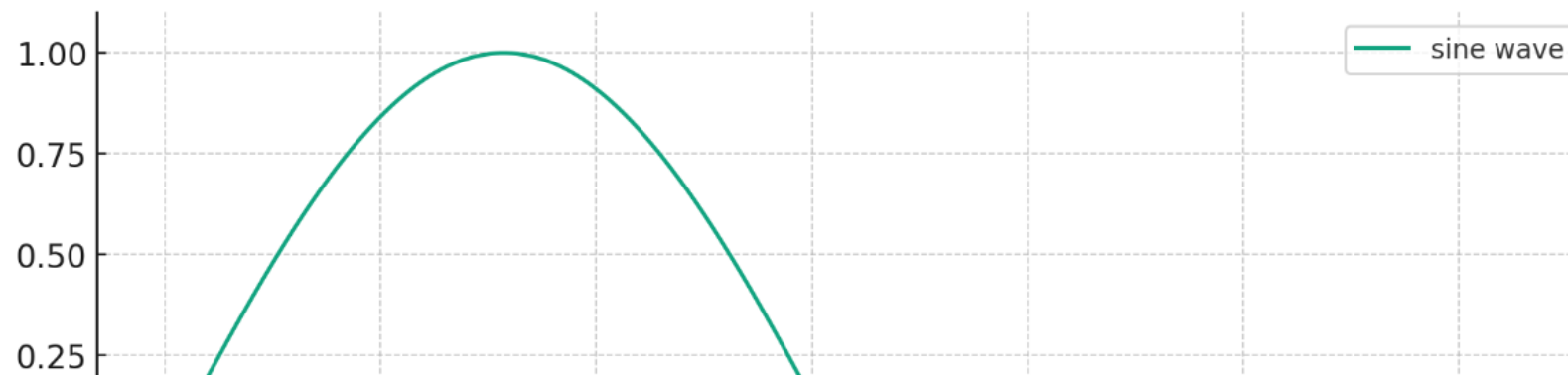
Assistants playground

Help

+ New assistant ↑ Select assistant 🗑 Delete </> View code

Show setup Clear chat Logs Thread files v ☐ JSON response ⓘ 608 tokens ⓘ

Plot of a Sine Wave



Type user query here. (Shift + Enter for new line)

Messages in the Assistants playground are visible to anyone with access to this resource and using the API.



Azure AI Foundry | Azure OpenAI Service

Chat

Assistants PREVIEW

Real-time audio PREVIEW

Images

Completions

Tools

Fine-tuning

Azure OpenAI PREVIEW

Evaluation

Batch jobs

Shared resources

Deployments

Quota

Safety + security

Data files

Vector stores PREVIEW

Assistants

+ New

Show

Type use

Sample Code

You can use the following code to start integrating your current prompt and settings into your application.

<https://pytechsummit2024demo.openai.azure.com/>

python

Entra ID authentication

Key authentication

Assistant setup

```
1 import os
2 import json
3 import requests
4 import time
5 from openai import AzureOpenAI
6
7 client = AzureOpenAI(
8     azure_endpoint = os.getenv("AZURE_OPENAI_ENDPOINT"),
9     api_key= os.getenv("AZURE_OPENAI_API_KEY"),
10     api_version="2024-05-01-preview"
11 )
12
13 assistant = client.beta.assistants.create(
```

Close

pytechsummit2024demo (eastus, S0)

Help

sine wave

to this resource and using the API.

EXPLORER

DEMO

math-tutor

assistant-math_tutor.ipynb

README.md

.env

requirements.txt

assistant-math_tutor.ipynb X

math-tutor > assistant-math_tutor.ipynb > ...

+ Code + Markdown | Run All Restart Clear All Outputs Variables Outline ...

```

process_message(
    """x=r*cos(u)sin(v)
y=r*sin(u)sin(v)
r=2+sin(7*u+5*v)

for 0<u<2π and 0<v<π.

Create a graph of the equation z=r*cos(v)."""
)

```

[8] ✓ 12.8s

... processing ...

Run completed successfully.

user:

```

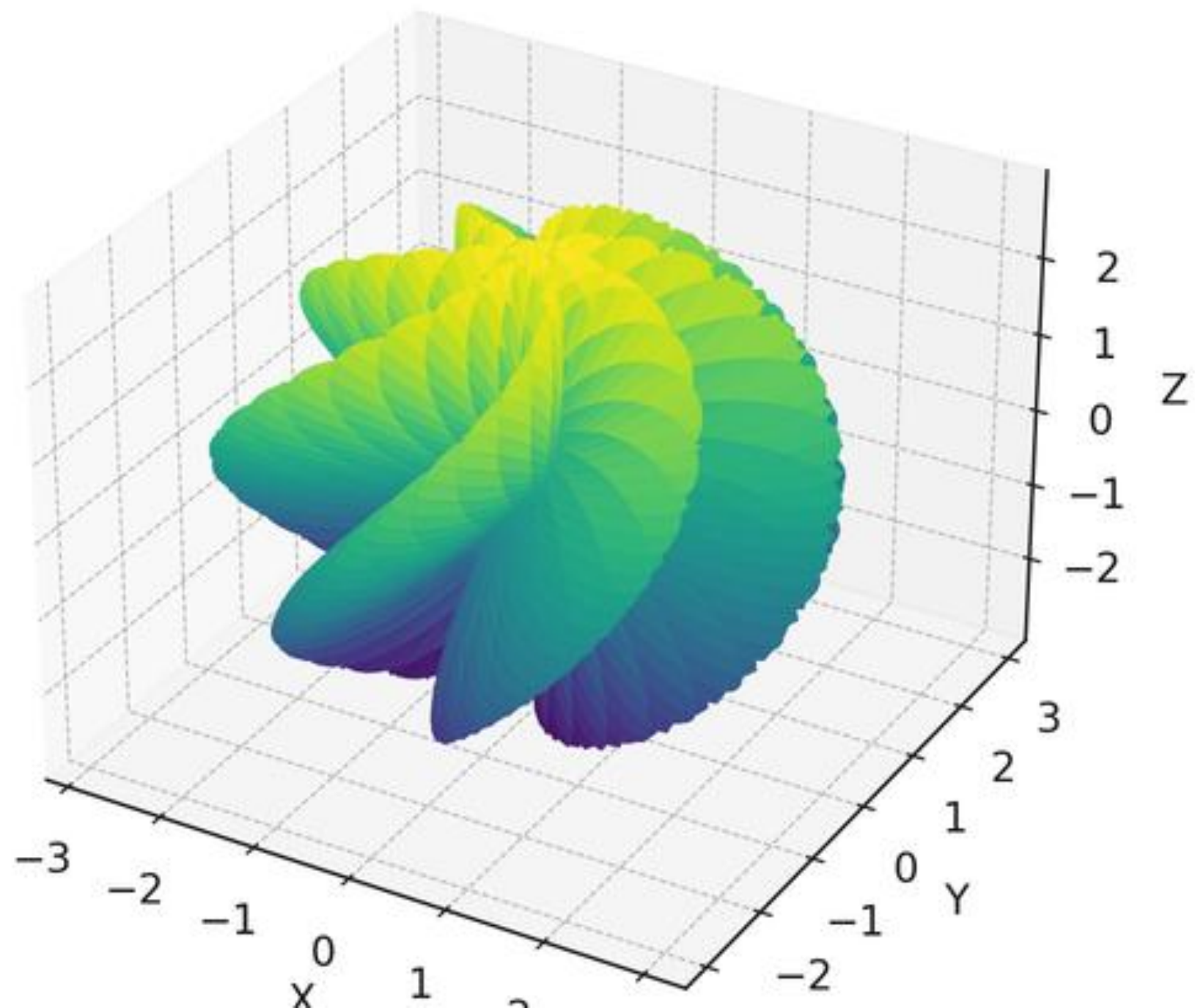
x=r*cos(u)sin(v)
y=r*sin(u)sin(v)
r=2+sin(7*u+5*v)

for 0<u<2π and 0<v<π.

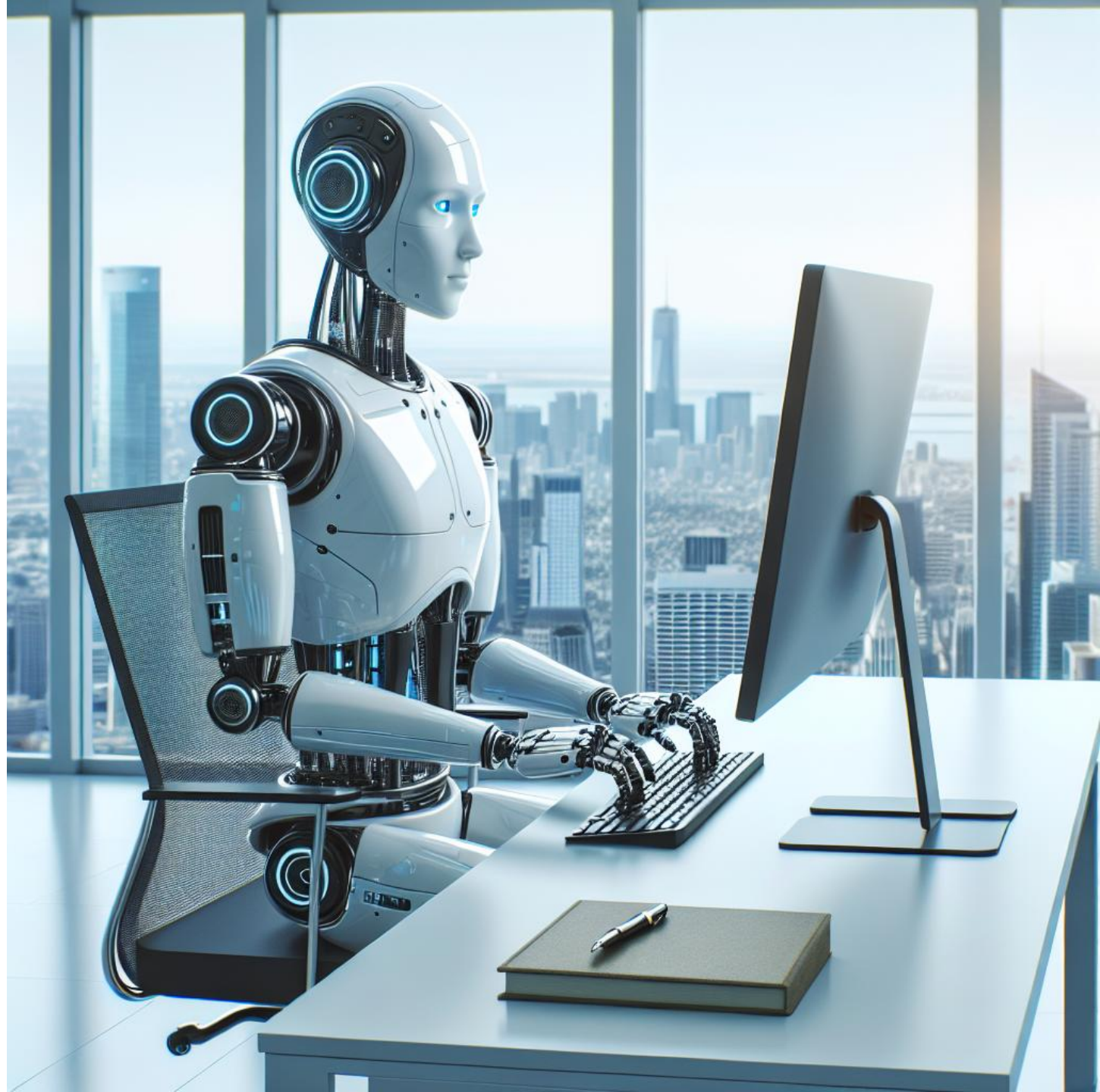
Create a graph of the equation z=r*cos(v).

assistant:
Let's create a 3D graph of the parametric surface defined by the equations:

```



Demo





Google Meet -
<https://meet.google.com/ffj-pdrc-bta>



Email: info@zalnet.pl

LinkedIn: <https://www.linkedin.com/in/beatazalewa/>

Blog: <https://zalnet.pl/blog>

Twitter: <https://x.com/beatazalewa>

GitHub: <https://github.com/beatazalewa/>

